1. Clinical Question & Core Hypothesis

Clinical Question

MMR-deficient (MSH6-altered/MSI-high) bladder tumors are expected to be immunogenic and responsive to immune checkpoint blockade, yet many do not benefit—why does high tumor neoantigen potential fail to translate into therapy response? 2) Tier 1 paths used: (a) MSH6 as an independent prognostic marker (mechanism: MMR → mutational/neoantigen load) and (b) MSI-driven AI biomarkers (mechanism: MSI/MMR context informs predictive signatures). I built on these because they directly link tumor-intrinsic MMR biology to expected immune responsiveness. 3) Tier 2 evidence that supports the claim: systemic immune status (LMR), hematologic ratios (HRR), circulating biomarkers (pZVR, pro-PSA), metabolic markers (GLU/microalbuminuria), and radiomic features can all modulate or report on immune competence and immune-exclusion phenotypes; these provide plausible modifiers that explain discordance between neoantigen load and clinical benefit. 4) Why therapy_response chosen as endpoint: MSH6/MSI biology is most immediately relevant to checkpoint inhibitor sensitivity in bladder cancer, so therapy_response is the clinically actionable outcome to explain and improve. 5) Deliberate exclusions: I excluded Tier 3 exploratory items (CSF-AD biomarkers, Gleason/post-surgical metrics, loosely linked PTEN narratives) to keep the hypothesis tightly focused on MMR-driven immunogenicity vs intratumoral immune infiltration; I also limited the core to only MSH6/MSI and one imaging feature to meet the Focusing Rule?

Working Hypothesis

In bladder cancers that are MSH6-altered/MSI-high, high neoantigen load creates a theoretical substrate for T cell–mediated anti-tumor activity, but actual response to single-agent PD-1/PD-L1 blockade is determined by (a) intratumoral immune contexture, specifically Tumor-Infiltrating Lymphocyte (TIL) density and spatial distribution (inflamed vs. immune-excluded phenotypes), and (b) systemic immune/metabolic state reflected by circulating and radiomic biomarkers. We hypothesize: (1) MSH6-altered/MSI-high tumors with high TIL density and inflamed spatial organization will have higher therapy_response rates to single-agent ICI compared with their low-TIL, immune-excluded counterparts, despite similar neoantigen loads. (2) Among MSH6-altered/MSI-high tumors with low TIL density, unfavorable systemic states (e.g., low LMR, high HRR, pro-inflammatory/catabolic metabolites, adverse radiomic signatures) will predict primary resistance to single-agent ICI. (3) Modulating either the TME (e.g., targeting TGF-β–driven exclusion, myeloid suppressive pathways) or the systemic milieu (e.g., metabolic/inflammatory modulation) will convert non-responders (low TIL and adverse systemic state) into responders, leading to improved outcomes with combination strategies. (4) An integrated biomarker model combining MSH6/MSI status, quantitative/ spatial TIL metrics, and a composite systemic-immune score will outperform neoantigen load alone in predicting therapy_response and guiding combination therapy decisions.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MSH6, MSI_status
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6-altered/MSI-high) bladder tumors are expected to be immunogenic and responsive to immune checkpoint blockade, yet many do not benefit—why does high tumor neoantigen potential fail to translate into therapy response? 2) Tier 1 paths used: (a) MSH6 as an independent prognostic marker (mechanism: MMR → mutational/neoantigen load) and (b) MSI-driven AI biomarkers (mechanism: MSI/MMR context informs predictive signatures).

MSI_status: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	LMR	Supporting modifier from knowledge graph
Secondary	pro-PSA	Supporting modifier from knowledge graph
Secondary	radiomics_features	Supporting modifier from knowledge graph
Secondary	HRR	Supporting modifier from knowledge graph
Secondary	pZVR	Supporting modifier from knowledge graph
Secondary	GLU	Supporting modifier from knowledge graph
Secondary	AI-based_biomarkers	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- LMR: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- pro-PSA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.